CWVS: Critical Window Variable Selection

CWVS_Example

[1] Simulate data for analysis:

- Setting the reproducibility seed and initializing packages for data simulation:

```
set.seed(4679)
library(CWVS)
library(boot)  #Inverse logit transformation
```

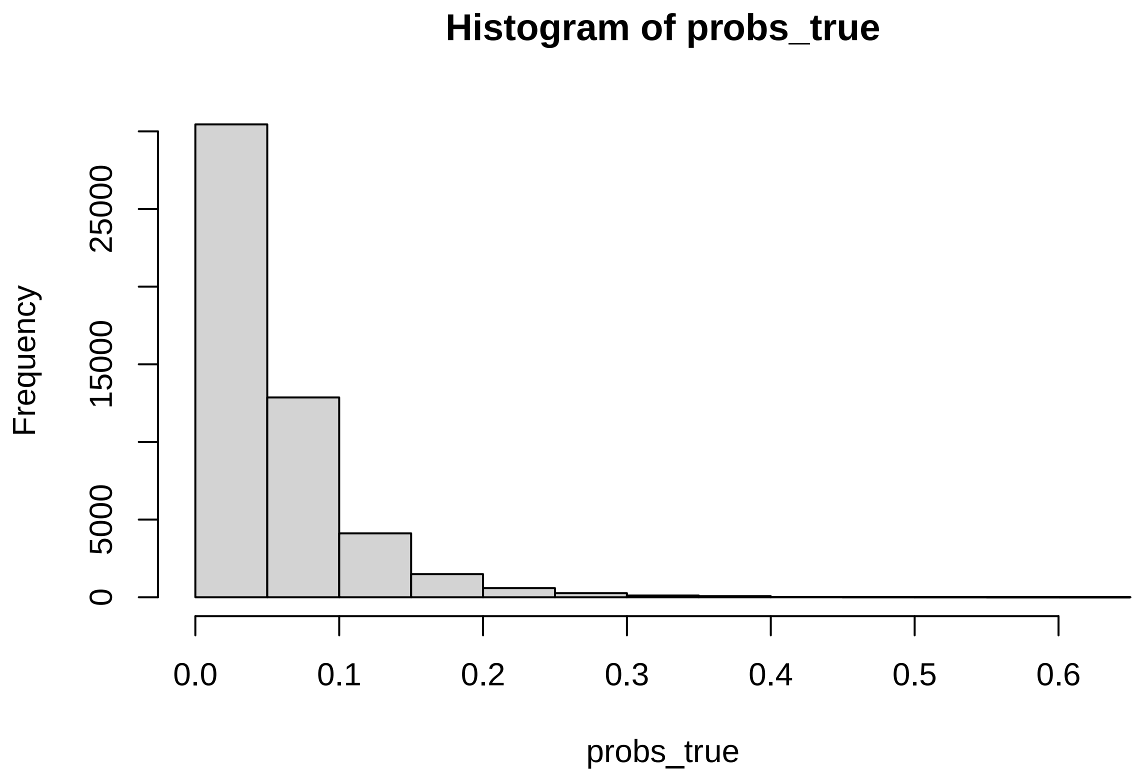
- Setting the global data values:

```
n<-50000  #Sample size
m<-27  #Number of exposure time periods
x<-matrix(1,
nrow=n,
ncol=2)  #Covariate design matrix
x[,2]<-scale(rnorm(n))

z<-matrix(rnorm(n=(n*m)),
nrow=n,
ncol=m)  #Exposure design matrix
for(j in 1:m){
z[,j]<-(z[,j] - median(z[,j]))/IQR(z[,j])  #Data standardization (interquartile range)
}
```

- Setting the values for the statistical model parameters:

```
beta_true<-c(-3.20, 0.10)
theta_true<-rep(0.60, times=m)
gamma_true<-c(rep(0, times=12),
rep(1, times=4),
rep(0, times=11))
alpha_true<-gamma_true*theta_true
logit_p_true<-x%*%beta_true +
z%*%alpha_true
probs_true<-inv.logit(logit_p_true)
hist(probs_true)
```




```
trials<-rep(1, times = n)
```

- Simulating the analysis dataset:

```
y<-rbinom(n=n,
          size=trials,
          prob=probs_true)
```

- Creating matched case-control data:

```
cases<-c(1:n)[y == 1]
controls<-c(1:n)[y == 0][1:length(cases)]
final_set<-c(cases,
             controls)
y<-y[final_set]

x<-x[final_set,]

z<-z[final_set,]

pair_id<-rep(c(1:length(cases)), times = 2)
```

[2] Fit CWVS to identify/estimate critical windows of susceptibility:

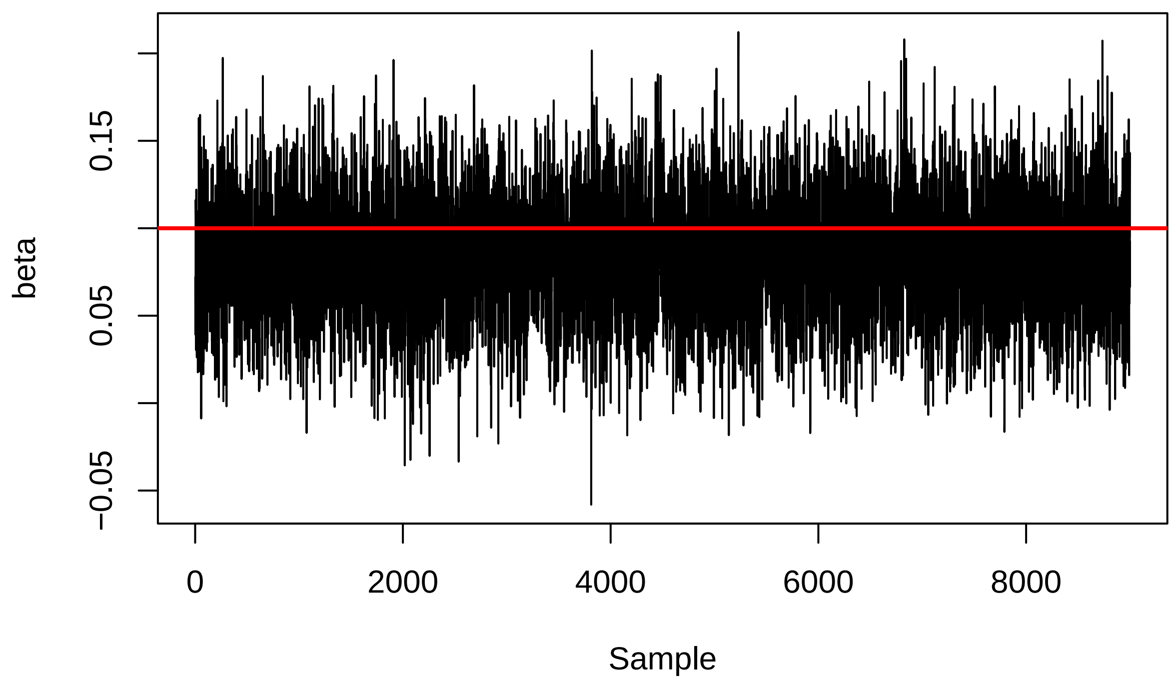
```
results<-CWVS(mcmc_samples = 10000,
              y = y, x = x, z = z,
              likelihood_indicator = 3,
              pair_id = pair_id,
```

```
             metrop_var_phi1_trans = 1.00,
             metrop_var_phi2_trans = 1.00,
             metrop_var_A11_trans = 0.03,
             metrop_var_A22_trans = 0.30)
```

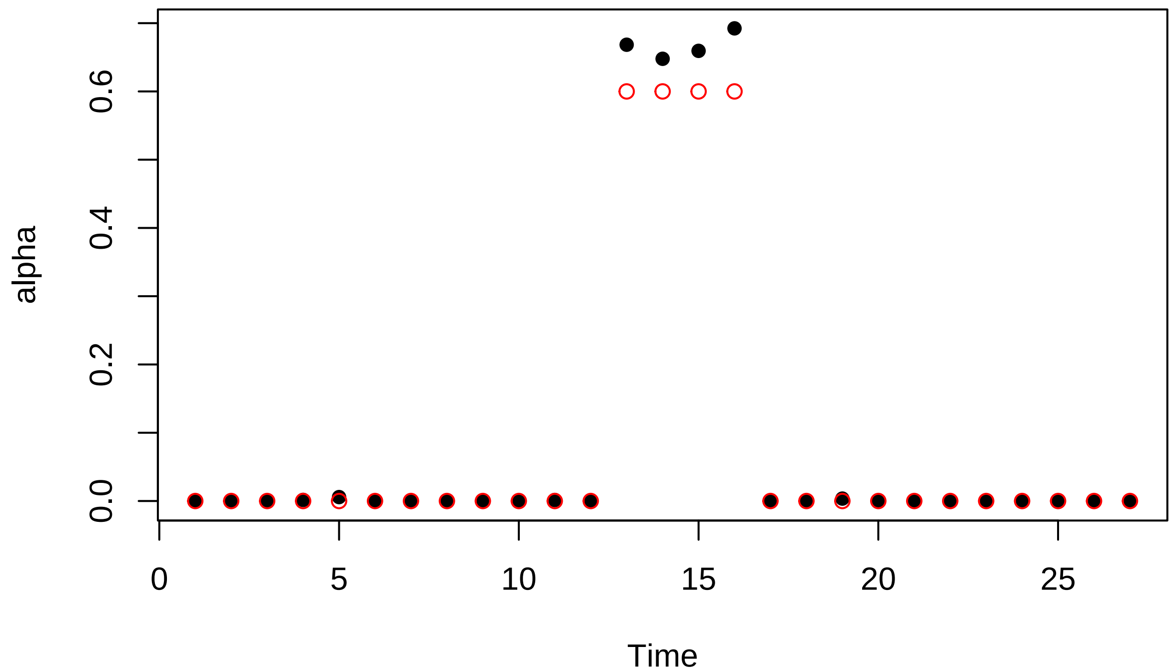
```
## Progress: 10%
## phi1 Acceptance: 37%
## phi2 Acceptance: 36%
## A11 Acceptance: 21%
## A22 Acceptance: 27%
## ********************
## Progress: 20%
## phi1 Acceptance: 37%
## phi2 Acceptance: 35%
## A11 Acceptance: 20%
## A22 Acceptance: 22%
## ********************
## Progress: 30%
## phi1 Acceptance: 38%
## phi2 Acceptance: 36%
## A11 Acceptance: 20%
## A22 Acceptance: 18%
## ********************
## Progress: 40%
## phi1 Acceptance: 37%
## phi2 Acceptance: 36%
## A11 Acceptance: 20%
## A22 Acceptance: 19%
## ********************
## Progress: 50%
## phi1 Acceptance: 37%
## phi2 Acceptance: 36%
## A11 Acceptance: 20%
## A22 Acceptance: 19%
## ********************
## Progress: 60%
## phi1 Acceptance: 36%
## phi2 Acceptance: 36%
## A11 Acceptance: 19%
## A22 Acceptance: 19%
## ********************
## Progress: 70%
## phi1 Acceptance: 36%
## phi2 Acceptance: 36%
## A11 Acceptance: 20%
## A22 Acceptance: 19%
## ********************
## Progress: 80%
## phi1 Acceptance: 36%
## phi2 Acceptance: 35%
## A11 Acceptance: 20%
## A22 Acceptance: 21%
## ********************
## Progress: 90%
```

```
## phi1 Acceptance: 36%
## phi2 Acceptance: 35%
## A11 Acceptance: 20%
## A22 Acceptance: 22%
## *******************
## Progress: 100%
## phi1 Acceptance: 36%
## phi2 Acceptance: 35%
## A11 Acceptance: 20%
## A22 Acceptance: 22%
## *******************
```

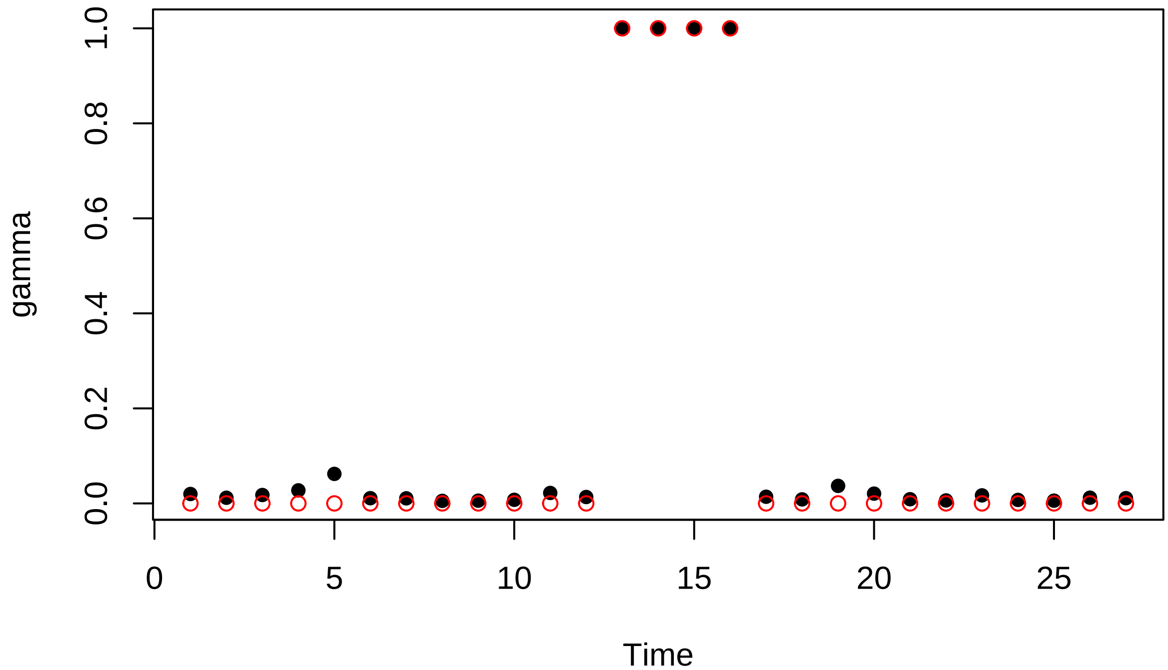
```r
plot(results$beta[1, 1001:10000],
     type="l",
     ylab="beta",
     xlab="Sample")
abline(h=beta_true[2],
       col="red",
       lwd=2)  #True value
```




```r
plot(rowMeans(results$alpha[,1001:10000]),
              pch=16,
              ylab="alpha",
              xlab="Time")
points(alpha_true,
       col="red")
```




```
plot(rowMeans(results$gamma[,1001:10000]),
     pch=16,
     ylab="gamma",
     xlab="Time")
points(gamma_true,
       col="red")
```


gamma
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25
Time